

A04 DATA WRANGLERS: DECODING THE DATA MAZE

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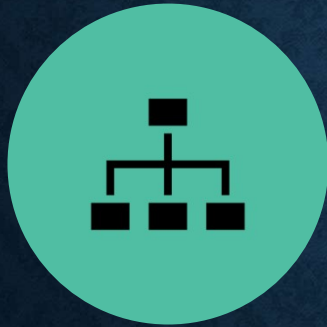
Richard Rodriguez

THE CHALLENGE

- A STREAMING PLATFORM AIMS TO FORECAST CUSTOMER CHURN
- THE DATASET CONTAINS INFORMATION SUCH AS AGES, GEOGRAPHIC LOCATIONS, SUBSCRIPTION PLANS, AND PAYMENT OPTIONS
- HOWEVER, THE DATA IS DISORGANIZED AND LACKS CONSISTENCY

24	24	Standard	UK	41	Ohio	Florida
50	19	Premium	Illinois	Monthly	Debit Card	33
41	Kansas	Basic		\$0.99	\$9.99	\$0.99
65	Debit Card	Credit	Ohio	33	Monthly	Monthly
30	USA	Card	Texas	\$9.99	Monthly	Florida
8	29	Texas	Stockholm	79	Ohio	Basic
29	42	Stockholm	Premium	Texas	Sweden	Sweden
58	Premium	Illinois	79	\$12.99	Kansas	22
27	New Jersey	UK	Texas	Kansas	Texas	Basic
37	New	Jexes	38	Kansas	Stockholm	Premium
37	Jersey	UK	Basic	Debit	Debit Card	Ireland

OUR DATA TYPE: CUSTOMER PROFILES



- STRUCTURED DATA



- CRITICAL FOR
UNDERSTANDING
CUSTOMER BEHAVIOR



- FOUNDATION FOR CHURN
PREDICTION

DATA QUALITY ISSUE #1: MISSING AGES

- Hypothetical: ~15% missing

- Younger users more likely to skip

- Creates demographic imbalance

**FIX #1:
MEDIAN
IMPUTATION +
MISSING
INDICATOR**

-
- Impute using median age by region
-
- Add binary 'age_missing' feature
-
- Reduces bias and preserves patterns

DATA QUALITY ISSUE #2: INCONSISTENT LOCATIONS



Hypothetical:
~20% inconsistent



“TX”, “Texas”,
“Tx.”



International
formats vary

FIX #2: STANDARDIZATION + GEOCODING

01

Convert to
consistent format

02

Optional: add
latitude/longitude

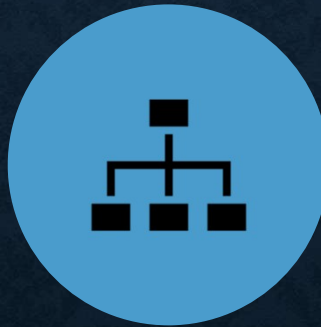
03

Encode with one
hot or
embeddings

DATA QUALITY ISSUE #3: SUBSCRIPTION TYPE VARIANTS



- HYPOTHETICAL: 10+ VARIANTS



- ONLY 3 REAL TIERS



- ARTIFICIAL CATEGORIES
CONFUSE THE MODEL

FIX #3: CATEGORY NORMALIZATION

- Map all variants to 3 tiers



- Reduces noise



- Improves interpretability

DATA QUALITY ISSUE #4: PAYMENT METHOD NOISE & DUPLICATES

Hypothetical: ~5–10% inconsistent

Duplicate customer records from payment changes

Causes data leakage

FIX #4: CLEANING + DEDUPLICATION



- Standardize payment labels via mapping



- Deduplicate: keep most recent record per customer



- Eliminates data leakage risk

DATA QUALITY ISSUE #5: IMPOSSIBLE AGES

Hypothetical:
~1% invalid

Ages like 3,
250, -7

Distorts
distribution

FIX #5: OUTLIER DETECTION



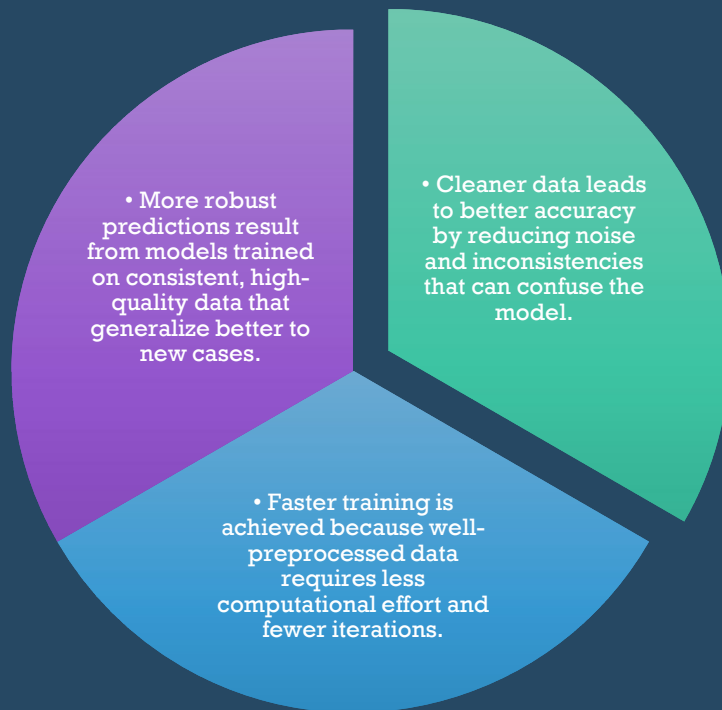
- Enforce valid age range



- Replace with median or 'unknown'



- Improves stability



WHY PREPROCESSING MATTERS

FINAL TAKEAWAYS

- Data wrangling is essential for transforming raw, messy data into a clean, usable format that models can learn from effectively.
- Even structured data can become messy due to inconsistencies, missing values, and errors that require careful handling.
- Preprocessing unlocks model performance by improving data quality, reducing noise, and enabling models to generalize better to new data.
- Investing time in preprocessing saves effort downstream by preventing model errors and improving prediction reliability.
- A robust preprocessing pipeline is a key foundation for successful machine learning projects.

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